Addendum: The Planck Radiation Law and the Origin of Number-Theoretic π

Addendum to: "The Geometric Vocabulary in Quantum Field Theory" and "Number-Theoretic π in the Casimir Effect"

Dat Nguyen Dtnguyen09@gmail.com ORCID: 0009-0001-2514-7768

Independent Researcher

---

1. Purpose

The Casimir analysis (Nguyen 2026d) identified three pathways by which π enters physics equations: geometric (angular integrals), statistical (Gaussian integrals), and number-theoretic (Euler's $\zeta(2k)$ formula). This addendum applies the classification to the Planck radiation law (Equation #48 in the catalog) and identifies a structural observation: the spectral and integrated forms of the same equation contain different types of π. Mode summation — integration over the thermal spectrum — is the physical operation that produces number-theoretic π.

---

2. The Spectral Form Contains Only Geometric π

The spectral energy density of a black body in D = 3 spatial dimensions is:

$u(\nu,T) = (8\pi h\nu^3/c^3) \times 1/(e^{(h\nu/kT)} - 1)$

The prefactor $8\pi\nu^2/c^3$ arises from counting electromagnetic modes in a cavity:

- $4\pi k^2\,dk$: the surface area of a spherical shell in k-space — $S(S^2)$, **Pathway 1** (geometric).
- $(2\pi)^3$: the volume of one mode cell in k-space — $C(\text{phase})^3$, **Pathway 1** (geometric).
- $k = 2\pi\nu/c$ and $dk/d\nu = 2\pi/c$: converting wavenumber to frequency — C(phase), **Pathway 1** (geometric).
- 2: transverse polarizations — **non-geometric** (degeneracy count, same as identified in the original catalog).

Every π in the spectral form traces to angular integration or phase-space periodicity. The Bose-Einstein occupation factor $1/(e^x - 1)$ contains no π. No Pathway 2 or Pathway 3 π is present.

Classification: All Type 1. Standard.

---

3. The Integrated Form Contains Both Geometric and Number-Theoretic π

Integrating over all frequencies to obtain the Stefan-Boltzmann energy density:

$u(T) = \int_0^\infty u(\nu,T)\, d\nu = (8\pi h/c^3) \times (kT/h)^4 \times \int_0^\infty x^3/(e^x - 1)\, dx$

The integral evaluates to $\Gamma(4)\zeta(4) = 6 \times \pi^4/90 = \pi^4/15$, giving:

$u(T) = \pi^2 k^4 T^4 / (15\hbar^3 c^3)$

The π^2 in this result is a net tally of two pathway contributions:

- **Pathway 1** (geometric): $S(S^2) = 4\pi$ contributes π^1 in the numerator; $h^3 = (2\pi\hbar)^3$ contributes π^3 in the denominator. Net Pathway 1: $\pi^{(1-3)} = \pi^{-2}$.
- **Pathway 3** (number-theoretic): $\zeta(4) = |B_4|(2\pi)^4/[2 \times 4!] = (2\pi)^4/1440$ contributes π^4, entering through the Bernoulli generating function.

Net: $\pi^{-2} \times \pi^4 = \pi^2$.

The factor 15 in the denominator decomposes as $90/\Gamma(4) = 90/6$, where $90 = \pi^4/\zeta(4)$ is the number-theoretic factor from the Bernoulli number $|B_4| = 1/30$, and $6 = 3!$ is combinatorial (from the Bose-Einstein integral).

Classification: Pathway 1 identifications are Standard. Pathway 3 identification of ζ(4) is Consistent with the Casimir analysis (Nguyen 2026d). The π tally decomposition is New.

---

4. Mode Summation as Type Conversion

The spectral form contains only Pathway 1 π. The integrated form contains both Pathway 1 and Pathway 3. The same equation, before and after integration, has different π-type content.

The operation that introduces Pathway 3 is the mode summation: $\int_0^\infty x^D/(e^x - 1)\, dx = \Gamma(D+1)\zeta(D+1)$. This integral is the continuous limit of the discrete mode sum $\Sigma\, n^{(-(D+1))}$, and it produces $\zeta(D+1)$. When D+1 is even (D odd), $\zeta(D+1)$ carries $(2\pi)^{(D+1)}$ through Euler's formula and introduces Pathway 3 π. When D+1 is odd (D even), $\zeta(D+1)$ evaluates to an irrational constant with no known π expression, and no Pathway 3 π appears.

The integrated Planck law thus follows the same even/odd boundary identified in the Casimir analysis: at odd D, the Stefan-Boltzmann coefficient decomposes fully into geometric and number-theoretic factors. At even D, it contains ζ(odd) with no geometric reading.

This is not a coincidence. Both the Casimir effect and the Planck law count modes of the same quantum field. The Casimir effect sums zero-point energies ($T = 0$, boundary-imposed discretization). The Planck law sums thermally occupied energies ($T > 0$, thermal occupation of continuum modes). Both produce $\zeta(D+1)$ because the mode-counting structure is shared. The vocabulary reads the same word — $\zeta(D+1)$ — in both contexts.

---

5. Summary

The Planck radiation law, read through the three-pathway classification, demonstrates that number-theoretic π (Pathway 3) is not a property of the spectral physics but a product of the mode summation. The spectral form contains only geometric π; integration over the thermal spectrum introduces $\zeta(4)$, carrying $(2\pi)^4$ from the Bernoulli generating function. The same $\zeta(D+1)$ appears in both the Planck and Casimir contexts because both sum modes of the same quantum field, and the same even/odd boundary applies.

References

- Casimir, H.B.G. (1948). "On the attraction between two perfectly conducting plates." Proceedings of the Royal Netherlands Academy of Arts and Sciences 51, 793–795.
- Cheng, H., Li, X.-L., and Liu, Y. (2010). "The Casimir force of Quantum Spring in the (D+1)-dimensional spacetime." arXiv:1008.3020.
- Nguyen, D. (2026a). "The Geometric Vocabulary of Fundamental Constants." Zenodo preprint.
- Nguyen, D. (2026b). "Reading the Equations: 55 Formulas in the Geometric Vocabulary." Zenodo preprint.
- Nguyen, D. (2026c). "The Geometric Vocabulary in Quantum Field Theory: Circumferences and Surfaces in One-Loop QED." Zenodo preprint.
- Nguyen, D. (2026d). "Number-Theoretic π in the Casimir Effect: A Third Pathway in the Geometric Vocabulary." Zenodo preprint.

Correspondence: Dat Nguyen, Dtnguyen09@gmail.com ORCID: 0009-0001-2514-7768